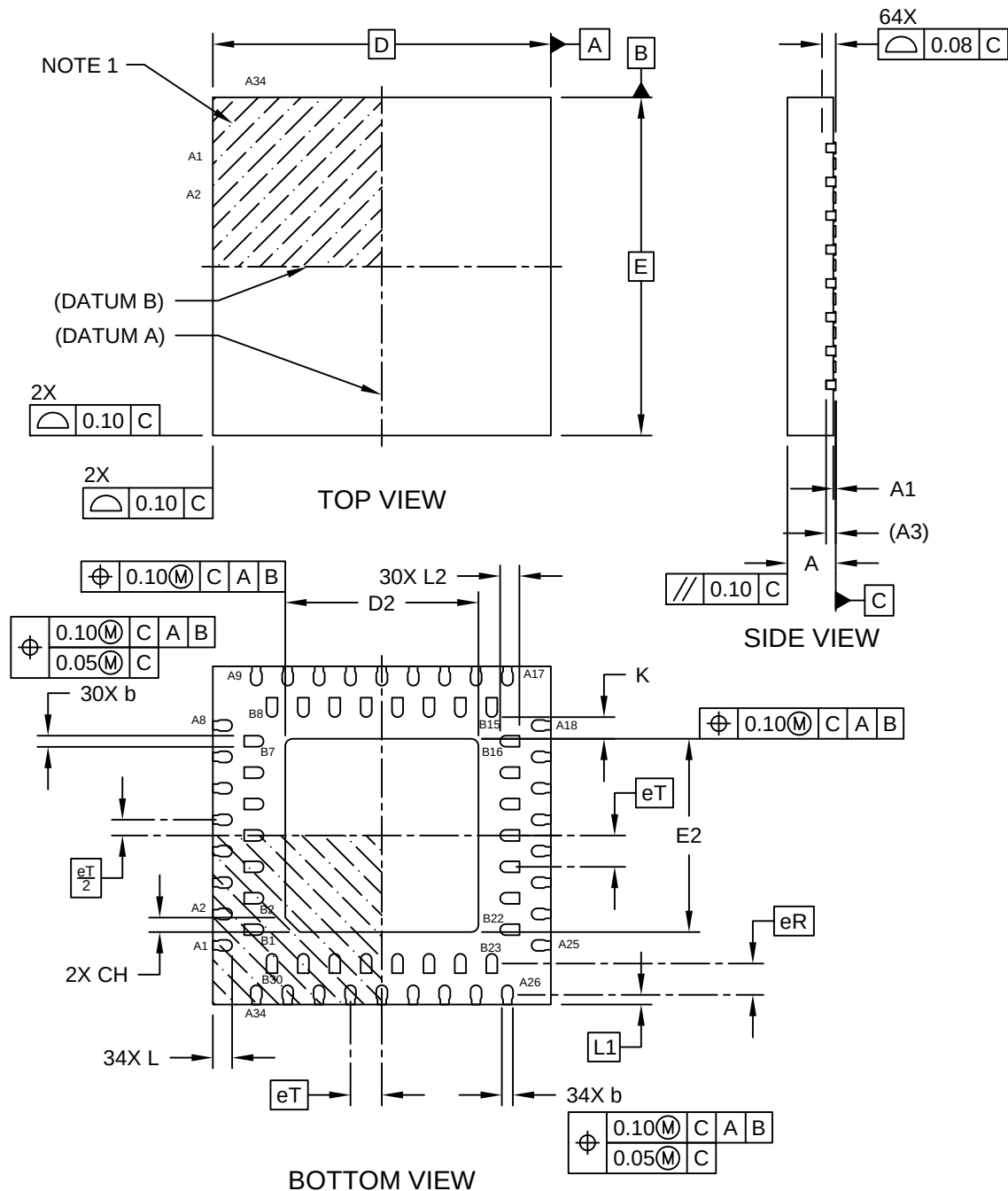


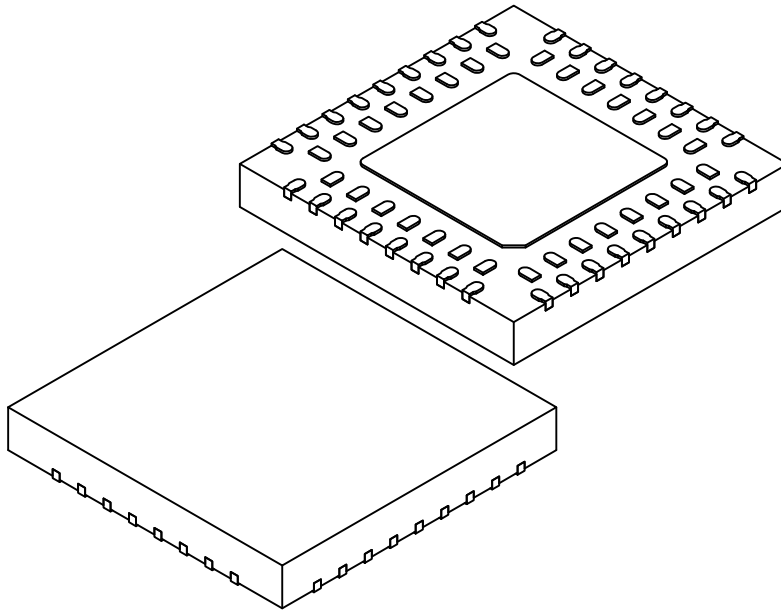
# **64-Lead Very Thin Plastic Quad Flat, No Lead Package (SUB) - 7x7x1.0 mm Body [VQFN]** **Dual Row; Atmel Legacy Global Package Code ZXC**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**64-Lead Very Thin Plastic Quad Flat, No Lead Package (SUB) - 7x7x1.0 mm Body [VQFN]  
Dual Row; Atmel Legacy Global Package Code ZXC**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



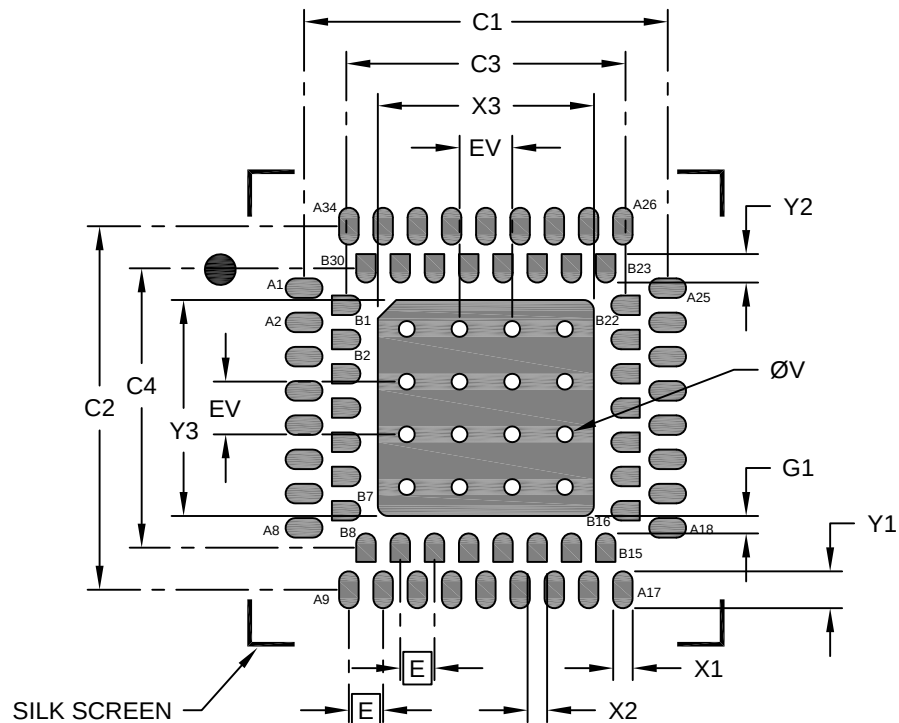
|                                      |    | Units | MILLIMETERS |      |      |
|--------------------------------------|----|-------|-------------|------|------|
| Dimension Limits                     |    |       | MIN         | NOM  | MAX  |
| Number of Terminals                  | N  |       | 64          |      |      |
| Pitch                                | eT |       | 0.65 BSC    |      |      |
| Outer Row Center to Inner Row Center | eR |       | 0.65 BSC    |      |      |
| Overall Height                       | A  |       | 0.80        | 0.85 | 0.90 |
| Standoff                             | A1 |       | 0.00        | 0.02 | 0.05 |
| Terminal Thickness                   | A3 |       | 0.20 REF    |      |      |
| Overall Length                       | D  |       | 7.00 BSC    |      |      |
| Exposed Pad Length                   | D2 |       | 3.90        | 4.00 | 4.10 |
| Overall Width                        | E  |       | 7.00 BSC    |      |      |
| Exposed Pad Width                    | E2 |       | 3.90        | 4.00 | 4.10 |
| Terminal Width                       | b  |       | 0.18        | 0.23 | 0.28 |
| Outer Row Terminal Length            | L  |       | 0.35        | 0.40 | 0.45 |
| Edge to Terminal Center              | L1 |       | 0.20 BSC    |      |      |
| Inner Row Terminal Length            | L2 |       | 0.35        | 0.40 | 0.45 |
| Terminal to Exposed-Pad              | K  |       | 0.20        | -    | -    |
| Exposed Pad Index Chamfer            | CH |       | -           | 0.30 | -    |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

# **64-Lead Very Thin Plastic Quad Flat, No Lead Package (SUB) - 7x7x1.0 mm Body [VQFN] Dual Row; Atmel Legacy Global Package Code ZXC**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## **RECOMMENDED LAND PATTERN**

| Units                           | MILLIMETERS |          |      |
|---------------------------------|-------------|----------|------|
|                                 | MIN         | NOM      | MAX  |
| Contact Pitch                   | E           | 0.65 BSC |      |
| Optional Center Pad Width       | X3          |          | 4.10 |
| Optional Center Pad Length      | Y3          |          | 4.10 |
| Outer Contact Pad Spacing       | C1          | 6.90     |      |
| Outer Contact Pad Spacing       | C2          | 6.90     |      |
| Inner Contact Pad Spacing       | C3          | 5.30     |      |
| Inner Contact Pad Spacing       | C4          | 5.30     |      |
| Outer Contact Pad Width (X34)   | X1          |          | 0.37 |
| Outer Contact Pad Length (X34)  | Y1          |          | 0.70 |
| Inner Contact Pad Width (X30)   | X2          |          | 0.37 |
| Inner Contact Pad Length (X30)  | Y2          |          | 0.54 |
| Contact Pad to Center Pad (X30) | G1          | 0.33     | 0.54 |
| Thermal Via Diameter            | V           | 0.30     |      |
| Thermal Via Pitch               | EV          | 1.00     |      |

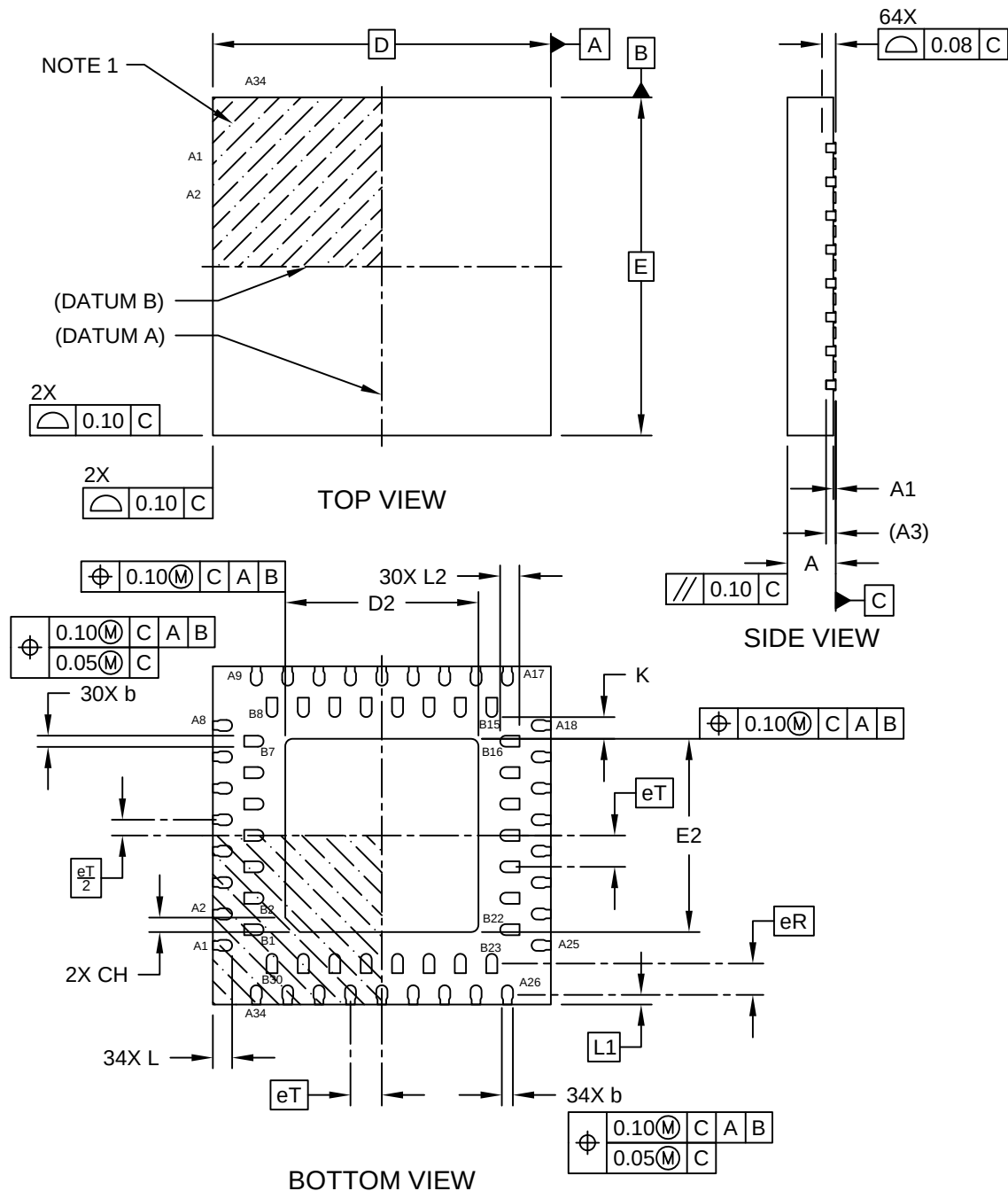
Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23420-SUB Rev A

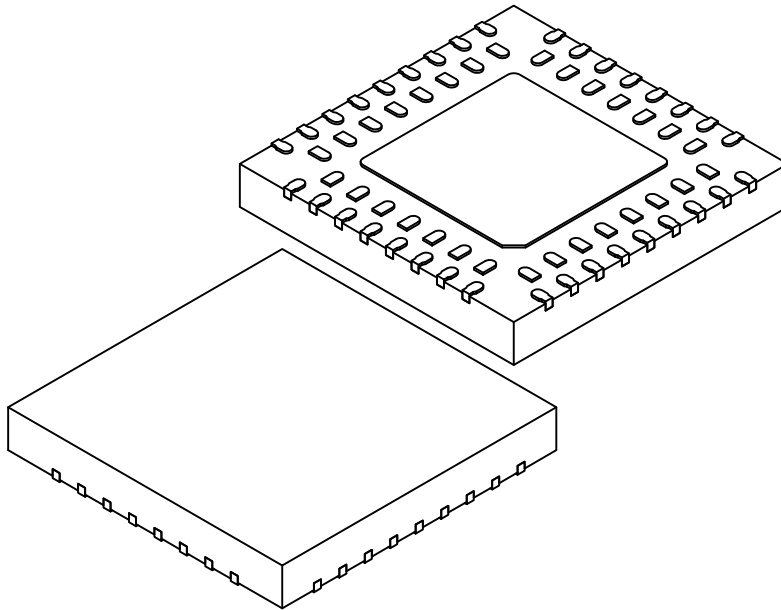
# **64-Lead Very Thin Plastic Quad Flat, No Lead Package (SVB) - 7x7x1.0 mm Body [VQFN]** **Dual Row; Atmel Legacy Global Package Code ZXC**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**64-Lead Very Thin Plastic Quad Flat, No Lead Package (SVB) - 7x7x1.0 mm Body [VQFN]  
Dual Row; Atmel Legacy Global Package Code ZXC**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



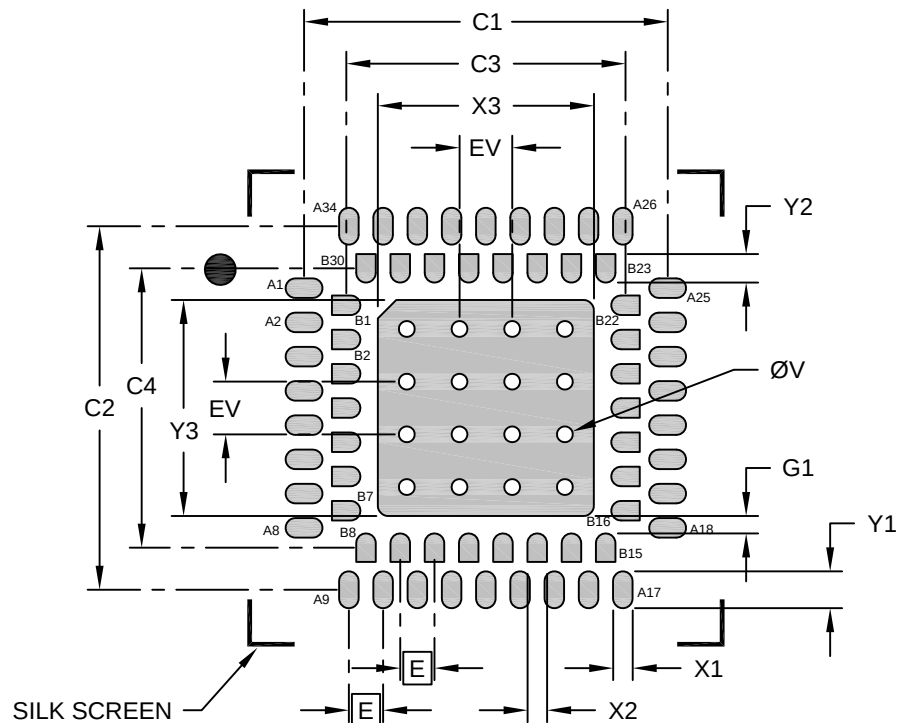
|                                      |    | Units | MILLIMETERS |      |      |
|--------------------------------------|----|-------|-------------|------|------|
| Dimension Limits                     |    |       | MIN         | NOM  | MAX  |
| Number of Terminals                  | N  |       | 64          |      |      |
| Pitch                                | eT |       | 0.65 BSC    |      |      |
| Outer Row Center to Inner Row Center | eR |       | 0.65 BSC    |      |      |
| Overall Height                       | A  |       | 0.80        | 0.85 | 0.90 |
| Standoff                             | A1 |       | 0.00        | 0.02 | 0.05 |
| Terminal Thickness                   | A3 |       | 0.20 REF    |      |      |
| Overall Length                       | D  |       | 7.00 BSC    |      |      |
| Exposed Pad Length                   | D2 |       | 3.90        | 4.00 | 4.10 |
| Overall Width                        | E  |       | 7.00 BSC    |      |      |
| Exposed Pad Width                    | E2 |       | 3.90        | 4.00 | 4.10 |
| Terminal Width                       | b  |       | 0.18        | 0.23 | 0.28 |
| Outer Row Terminal Length            | L  |       | 0.35        | 0.40 | 0.45 |
| Edge to Terminal Center              | L1 |       | 0.20 BSC    |      |      |
| Inner Row Terminal Length            | L2 |       | 0.35        | 0.40 | 0.45 |
| Terminal to Exposed-Pad              | K  |       | 0.20        | -    | -    |
| Exposed Pad Index Chamfer            | CH |       | -           | 0.30 | -    |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

# 64-Lead Very Thin Plastic Quad Flat, No Lead Package (SVB) - 7x7x1.0 mm Body [VQFN] Dual Row; Atmel Legacy Global Package Code ZXC

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## RECOMMENDED LAND PATTERN

| Dimension Limits                | Units | MILLIMETERS |      |      |
|---------------------------------|-------|-------------|------|------|
|                                 |       | MIN         | NOM  | MAX  |
| Contact Pitch                   | E     | 0.65 BSC    |      |      |
| Optional Center Pad Width       | X3    |             |      | 4.10 |
| Optional Center Pad Length      | Y3    |             |      | 4.10 |
| Outer Contact Pad Spacing       | C1    |             | 6.90 |      |
| Outer Contact Pad Spacing       | C2    |             | 6.90 |      |
| Inner Contact Pad Spacing       | C3    |             | 5.30 |      |
| Inner Contact Pad Spacing       | C4    |             | 5.30 |      |
| Outer Contact Pad Width (X34)   | X1    |             |      | 0.37 |
| Outer Contact Pad Length (X34)  | Y1    |             |      | 0.70 |
| Inner Contact Pad Width (X30)   | X2    |             |      | 0.37 |
| Inner Contact Pad Length (X30)  | Y2    |             |      | 0.54 |
| Contact Pad to Center Pad (X30) | G1    | 0.33        |      | 0.54 |
| Thermal Via Diameter            | V     |             | 0.30 |      |
| Thermal Via Pitch               | EV    |             | 1.00 |      |

Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23420-SVB Rev A